nbzinbKoo: Negative Binomial and Zero-Inflated Negative Binomial Regression by IRLS and EM

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1 Introduction

Counts such as the number of doctor visits are often **overdispersed**: their variance is larger than their mean, which the Poisson model does not allow. They often also contain **more zeros** than a Poisson (or even a negative binomial) distribution predicts. The negative binomial (NB2) model handles overdispersion through a dispersion parameter θ. The zero-inflated negative binomial (ZINB) model also allows a separate process that produces "structural" zeros.

The **nbzinbKoo** package fits both models with algorithms written from scratch: iteratively reweighted least squares (IRLS) alternated with Newton–Raphson for NB regression, and an EM algorithm whose M-steps are IRLS fits for ZINB regression. It does not call `optim()` or any other general-purpose optimizer. This report describes the methods (Section 2) and the package (Section 3), checks the results against `MASS::glm.nb()` and `pscl::zeroinfl()` (Section 4), studies the ZINB estimator by simulation (Section 5), and analyzes physician office visits in the NMES1988 data (Section 6).

2 Methods

2.1 Negative binomial regression

For counts y_i with covariates x_i and offset o_i, the NB2 model has $\log \mu_i = \eta_i = x_i^\top \beta + o_i$ and $\mathrm{Var}(Y_i) = \mu_i + \mu_i^2/\theta$. With prior weights a_i (all 1 in an ordinary fit) the log-likelihood is

$$\ell(\beta, \theta) = \sum_i a_i \Big[\log \Gamma(y_i + \theta) - \log \Gamma(\theta) - \log y_i! + \theta \log \theta + y_i \log \mu_i - (\theta + y_i) \log(\theta + \mu_i) \Big].$$

IRLS for β (θ fixed). Fisher scoring for a log-link GLM with variance $V(\mu) = \mu + \mu^2/\theta$ is weighted least squares with

$$w_i = a_i \frac{(\partial \mu_i / \partial \eta_i)^2}{V(\mu_i)} = a_i \frac{\theta \mu_i}{\theta + \mu_i}, \qquad z_i = (\eta_i - o_i) + \frac{y_i - \mu_i}{\mu_i}, \qquad \beta^{(t+1)} = (X^\top W X)^{-1} X^\top W z.$$

The least-squares step is solved by a QR decomposition of $W^{1/2}X$, and a step that lowers ℓ is halved.

Newton–Raphson for θ (β fixed). With $U(\theta) = \sum_i a_i[\psi(y_i+\theta) - \psi(\theta) + \log\theta + 1 - \log(\theta+\mu_i) - (y_i+\theta)/(\theta+\mu_i)]$ and $H(\theta) = \sum_i a_i[\psi'(y_i+\theta) - \psi'(\theta) + 1/\theta - 2/(\theta+\mu_i) + (y_i+\theta)/(\theta+\mu_i)^2]$, we update $\phi = \log\theta$, which keeps $\theta > 0$: $\phi \leftarrow \phi - U_\phi/H_\phi$ with $U_\phi = \theta U$ and $H_\phi = \theta^2 H + \theta U$.

Outer loop and standard errors. We start from a Poisson IRLS fit and the moment estimate $\theta^{(0)} = \sum_i a_i / \sum_i a_i(y_i/\mu_i - 1)^2$, then alternate the two steps until ℓ changes by less than 10^{-8}. At convergence $\widehat{\mathrm{Var}}(\hat\beta) = (X^\top \hat W X)^{-1}$ (treating θ as known, as `glm.nb()` does) and $\mathrm{SE}(\hat\theta) = \{-H(\hat\theta)\}^{-1/2}$.

2.2 Zero-inflated negative binomial regression

The ZINB model mixes a point mass at zero (probability π_i, $\mathrm{logit}\,\pi_i = z_i^\top\gamma$) with an $\mathrm{NB}(\mu_i,\theta)$ count distribution. Its observed-data log-likelihood is

$$\ell(\beta,\gamma,\theta) = \sum_{y_i=0} \log\left[\pi_i + (1-\pi_i) f_{NB}(0;\mu_i,\theta)\right] + \sum_{y_i>0} \left[\log(1-\pi_i) + \log f_{NB}(y_i;\mu_i,\theta)\right],$$

with $f_{NB}(0;\mu,\theta) = \{\theta/(\theta+\mu)\}^\theta$. Because of the log of a sum, this is not an exponential-family likelihood and IRLS cannot be applied to it directly. We maximize it by EM, treating the indicator D_i of a structural zero as missing data.

- **E-step.** $\tau_i = P(D_i = 1 \mid y_i) = \pi_i / \{\pi_i + (1-\pi_i) f_{NB}(0;\mu_i,\theta)\}$ if $y_i = 0$ and $\tau_i = 0$ otherwise.
- **M-step, zero part.** Maximize $\sum_i [\tau_i \log\pi_i + (1-\tau_i)\log(1-\pi_i)]$: a logistic regression with fractional response τ_i, fitted by IRLS with $w_i = \pi_i(1-\pi_i)$ and $z_i = z_i^\top\gamma + (\tau_i - \pi_i)/\{\pi_i(1-\pi_i)\}$.
- **M-step, count part.** Maximize $\sum_i (1-\tau_i)\log f_{NB}(y_i;\mu_i,\theta)$: exactly the NB fit above with prior weights $a_i = 1 - \tau_i$ (the package reuses the same code).

EM starts from the NB fit for (β,θ) and a logistic IRLS fit of $I(y_i = 0)$ on z_i for γ, and stops when ℓ changes by less than 10^{-8}. The package records ℓ at every iteration and warns if it ever decreases.

Standard errors. We invert the observed information of ℓ in $(\beta,\gamma,\phi = \log\theta)$, computed **analytically**. For observation i let $h_i = \log f_{NB}(y_i;\mu_i,\theta)$ and $\rho_i = 1 - \tau_i$. Then

$$\frac{\partial \ell_i}{\partial \eta_z} = \tau_i - \pi_i, \quad \frac{\partial \ell_i}{\partial a} = \rho_i h_a, \quad \frac{\partial^2 \ell_i}{\partial \eta_z^2} = \tau_i(1-\tau_i) - \pi_i(1-\pi_i), \quad \frac{\partial^2 \ell_i}{\partial \eta_z\,\partial a} = -\tau_i\rho_i h_a, \quad \frac{\partial^2 \ell_i}{\partial a\,\partial b} = \rho_i h_{ab} + \tau_i\rho_i h_a h_b,$$

for $a, b \in \{\eta_c, \phi\}$, where $h_\eta = \theta(y-\mu)/(\theta+\mu)$, $h_{\eta\eta} = -\theta\mu(\theta+y)/(\theta+\mu)^2$, $h_\phi = \theta U_i$, $h_{\phi\phi} = \theta^2 H_i + \theta U_i$ and $h_{\eta\phi} = \theta\mu(y-\mu)/(\theta+\mu)^2$. The chain rule ($\partial\eta_c/\partial\beta = x_i$, $\partial\eta_z/\partial\gamma = z_i$) gives the blocks of the Hessian. The test suite checks this Hessian against `numDeriv::hessian()` of the package's own log-likelihood.

3 The R package

The package is installed with `devtools::install_github("gracko95/nbzinbKoo")`. Its main functions are:

Function	Purpose
`nbreg(formula, data, offset, control)`	NB2 regression (IRLS + Newton–Raphson)
`zinbreg(formula, data, offset, control)`	ZINB regression (EM with IRLS M-steps); formula `y ~ x | z`
`nbreg_control()`, `zinbreg_control()`	tolerances, maximum iterations, `trace`, Hessian type

Both functions return S3 objects (classes `"nbreg"` and `"zinbreg"`) with methods `print`, `summary`, `coef`, `vcov`, `logLik` (so `AIC()` and `BIC()` work), `fitted`, `residuals` (response and Pearson), `nobs` and `predict` (with `newdata`; for `zinbreg`, `type = "response"`, `"count"`, `"zero"` or `"prob0"`). The code is organized as `R/nbreg.R` (NB engine), `R/zinbreg.R` (EM engine and analytic Hessian), `R/methods.R` and `R/utils.R` (log-likelihood, θ derivatives, weighted least squares).

```
fit <- zinbreg(art ~ fem + kid5 + ment | ment, data = bioChemists)
round(coef(fit), 3)
```

```
count_(Intercept)    count_femWomen        count_kid5        count_ment
            0.494            -0.228            -0.131             0.024
 zero_(Intercept)         zero_ment
           -0.795            -0.619
```

```
c(theta = fit$theta, logLik = fit$loglik, EM_iterations = fit$iter)
```

```
       theta        logLik EM_iterations
    2.695777 -1554.752519    113.000000
```

4 Benchmarking

We compare `nbreg()` with `MASS::glm.nb()` and `zinbreg()` with `pscl::zeroinfl(dist = "negbin")` on `bioChemists` (Long, 1997) and `NMES1988` (Deb and Trivedi, 1997). The reference functions were run with tight tolerances (`epsilon = 1e-10` and `reltol = 1e-12`). Table 1 gives the maximum absolute differences, which are all far below the required 10^{-5} (NB coefficients and SEs), 10^{-4} (NB θ and log-likelihood) and 10^{-3} (ZINB). A positive log-likelihood difference means that our fit attains a slightly higher maximum.

Table 1: Maximum absolute differences from the reference implementations, iterations of our algorithm (outer NB iterations or EM iterations), and median elapsed time in seconds. For ZINB the theta row is log(theta).

Fit	Coef	SE	theta	logLik	Iter	Time (ours)	Time (ref.)
NB, bioChemists	2.1e-08	9.8e-10	6.7e-08	4.5e-13	4	0.015	0.018
NB, NMES1988	4.7e-08	9.2e-10	4.3e-08	7.1e-11	3	0.077	0.066
ZINB, bioChemists	1.5e-04	9.4e-05	1.3e-05	-4.2e-08	111	1.072	0.043
ZINB, NMES1988	2.6e-05	1.7e-05	4.4e-06	-1.6e-08	53	3.039	0.195

Table 2: ZINB estimates and standard errors for NMES1988: zinbreg() versus zeroinfl().

Term	zinbreg	zeroinfl	SE zinbreg	SE zeroinfl
count_(Intercept)	1.19372	1.19371	0.05666	0.05666
count_hospital	0.20148	0.20148	0.02036	0.02036
count_healthpoor	0.28513	0.28513	0.04509	0.04509
count_healthexcellent	-0.31934	-0.31934	0.06040	0.06040
count_chronic	0.12900	0.12900	0.01193	0.01193
count_gendermale	-0.08028	-0.08028	0.03102	0.03102
count_school	0.02142	0.02142	0.00436	0.00436
count_insuranceyes	0.12586	0.12586	0.04159	0.04159
zero_(Intercept)	-0.04689	-0.04688	0.26856	0.26856
zero_hospital	-0.80046	-0.80048	0.42080	0.42082
zero_chronic	-1.24791	-1.24794	0.17831	0.17832
zero_gendermale	0.64768	0.64768	0.20011	0.20011
zero_school	-0.08377	-0.08377	0.02625	0.02625
zero_insuranceyes	-1.17561	-1.17562	0.22012	0.22012
log(theta)	0.39415	0.39414	0.03504	0.03504

The side-by-side tables for the other three fits are produced by `inst/benchmark/benchmark.R` and look the same. EM needs more iterations than BFGS and is slower (each M-step is itself an iterative fit), but every run converged and the log-likelihood never decreased.

5 Simulation study

We simulated $R = 200$ data sets of size $n = 500$ (seed 651) from a ZINB model with $\log \mu_i = 1 + 0.5x_{1i} - 0.7x_{2i}$ ($x_1 \sim N(0,1)$, $x_2 \sim$ Bernoulli(0.5)), $\operatorname{logit} \pi_i = -1 + 0.8z_{1i}$ ($z_1 \sim N(0,1)$) and $\theta = 1.5$, and fitted each with `zinbreg()` (script `inst/benchmark/simulation.R`). 200 of 200 fits converged.

Table 3: Simulation results: bias, mean estimated SE versus Monte Carlo SD of the estimates, and coverage of 95% Wald intervals.

Parameter	True	Mean est.	Bias	Mean SE	MC SD	Coverage
count_(Intercept)	1.000	0.997	-0.003	0.099	0.101	0.930
count_x1	0.500	0.502	0.002	0.064	0.064	0.960
count_x2	-0.700	-0.709	-0.009	0.126	0.131	0.945
zero_(Intercept)	-1.000	-1.053	-0.053	0.329	0.335	0.960
zero_z1	0.800	0.840	0.040	0.235	0.246	0.955
log(theta)	0.405	0.444	0.039	0.252	0.248	0.965

TODO (your words): comment on the bias (is it small relative to the SE?), whether the mean SE matches the Monte Carlo SD (are the analytic SEs right?), and whether coverage is close to 0.95 given the Monte Carlo error $\sqrt{0.95 \cdot 0.05/R} \approx 0.015$.

6 Data analysis: physician office visits (NMES1988)

The NMES1988 data record the number of physician office visits (`visits`) for 4406 people aged 66 and over. The mean is 5.77 but the variance is 45.7, about 8 times larger, which is strong evidence of overdispersion. 15.5% of the people made no visits, while a Poisson distribution with the same mean predicts only 0.31% (Figure 1).

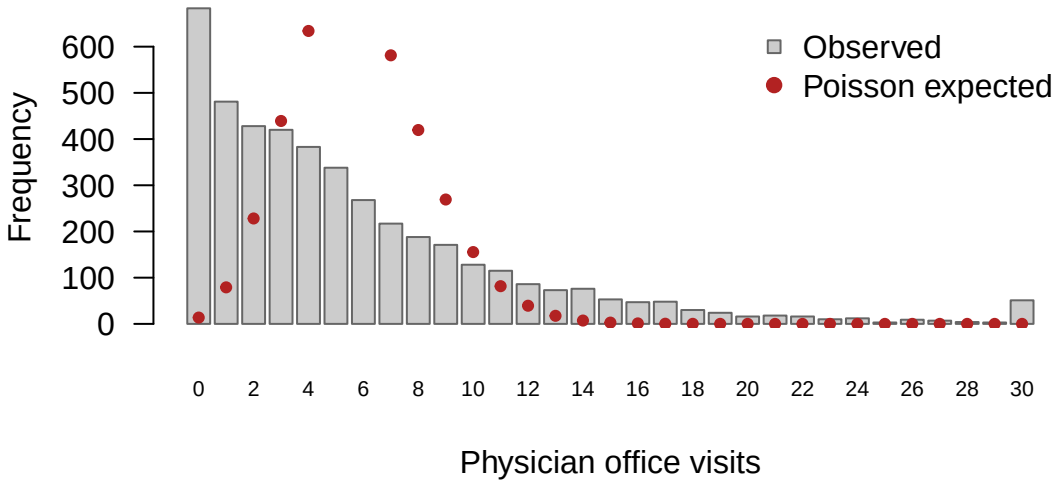


Figure 1: Observed distribution of visits (bars, 30 = 30 or more) and frequencies expected under a Poisson distribution with the same mean (points).

```r
f_count <- visits ~ hospital + health + chronic + gender + school + insurance
m_pois <- glm(f_count, family = poisson, data = NMES1988)      # reference only
m_nb   <- nbreg(f_count, data = NMES1988)
m_zinb <- zinbreg(visits ~ hospital + health + chronic + gender + school + insurance |
                    hospital + chronic + gender + school + insurance, data = NMES1988)
```

Table 4: Model comparison for NMES1988, including the expected number of zeros under each model.

Model	df	logLik	AIC	BIC	Expected zeros
Poisson	8	-17971.6	35959.2	36010.4	46.7
NB	9	-12170.6	24359.1	24416.6	608.0
ZINB	15	-12090.7	24211.4	24307.3	708.9

The data contain 683 zeros.

Table 5: Count part: incidence rate ratios (IRR) with 95% Wald confidence intervals.

Covariate	NB IRR (95% CI)	ZINB count IRR (95% CI)
hospital	1.24 (1.20, 1.29)	1.22 (1.18, 1.27)
healthpoor	1.36 (1.23, 1.49)	1.33 (1.22, 1.45)
healthexcellent	0.71 (0.63, 0.80)	0.73 (0.65, 0.82)
chronic	1.19 (1.16, 1.22)	1.14 (1.11, 1.16)
gendermale	0.88 (0.83, 0.94)	0.92 (0.87, 0.98)
school	1.03 (1.02, 1.04)	1.02 (1.01, 1.03)
insuranceyes	1.25 (1.16, 1.35)	1.13 (1.05, 1.23)

Table 6: ZINB zero part: odds ratios (OR) of being a structural zero (a person who would not visit a physician), with 95% Wald confidence intervals.

Covariate	OR (95% CI)	p-value
hospital	0.45 (0.20, 1.02)	0.0571
chronic	0.29 (0.20, 0.41)	<1e-04
gendermale	1.91 (1.29, 2.83)	0.0012
school	0.92 (0.87, 0.97)	0.0014
insuranceyes	0.31 (0.20, 0.48)	<1e-04

The NB model estimates $\hat{\theta} = 1.21$ (SE 0.034), and the ZINB model $\hat{\theta} = 1.48$.

TODO (your words; this is graded on interpretation, do not copy Appendix A):

1. What the descriptive summary and Table 4 say about overdispersion and excess zeros (compare the expected zeros of the NB model with the 683 observed).

2. Interpret the main IRRs in plain language (e.g. "each additional chronic condition multiplies the expected number of visits by …").
3. Interpret the zero-part ORs: which covariates change the odds of being a person who would never visit a physician, and in which direction?
4. Which model you prefer (AIC vs BIC, expected zeros, interpretability) and why.

7 Discussion

TODO (your words): limitations and extensions. Ideas to consider: EM is slower than BFGS (Table 1) and could be accelerated (e.g. a final few Newton steps on the full likelihood, or SQUAREM); the NB standard errors treat θ as known, as `glm.nb()` does; only the count part accepts an offset; other links for the zero part, zero-inflated Poisson or hurdle models, and robust (sandwich) standard errors are natural extensions.

8 AI-use statement

TODO (your words): describe how you used Claude Code (what you asked it to do), what you verified yourself (e.g. comparing the IRLS weights, τ_i and the θ score with Section 3, running the tests, the prohibited-function grep), and one concrete example of output you corrected or rejected.

9 References

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